

Nepal airborne gravity and geoid project.

Data acquisition and gravity processing report.

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Copenhagen October 24, 2011

Content

1. Overview of the airborne operations	p 1
2. Survey equipment	p 2
2. GPS processing and reference station coordinates.	p 3
3. Gravity ties	p 3
4. Airborne gravity processing	p 3
5. Data validation	p 4
6. Summary	p 5
Appendix A. Format of data file	p 6
Appendix B. Gravity reference stations	p 7

Overview of the airborne operations

The survey was flown with a COWI owned and operated Beechcraft 200 reg OY-CKP. The aircraft arrived in Kathmandu on November 27, 2010. The pilots underwent a training course at Kathmandu Flight Academy during the first week in Kathmandu in order to familiarize them with flight conditions in the mountains. Airborne gravity operations commenced on December 4 after the pilot's successful completion of the training course. The last flight took place on December 17. Altogether 13 flights took place during this period. Details of the flights are summarized in Table 1.

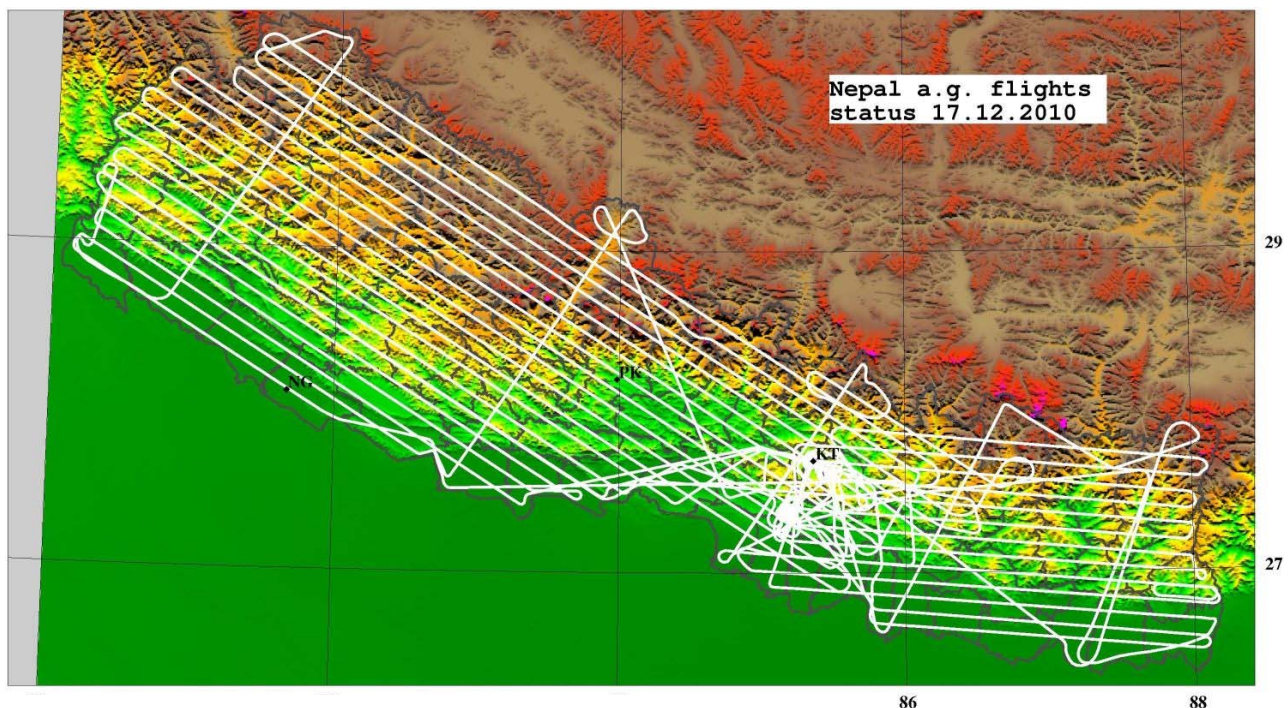


Figure 1. Ground track plot including all flights

Table 1: OY-CKP flights (time is UTC)

Date/JD	Lines	Airborne	Landed	Airborne time	Airborne time to date
04 Dec / 338	M2G1, N1N2, G2G3, H3H2	0358	0845	4h47	
05 Dec / 339	U1U2, T2T1	0909	1128	2h19	7h06
06 Dec / 340	K2K1, L1L2, L2L3, K3K2	0236	0857	6h21	13h27
07 Dec / 341	R2R1, Q1Q2, P2P1, O1O2	0249	0850	6h01	19h28
08 Dec / 342	J2J3, I3I2,	0331	0545	2h14	21h42
---	J2J1, I1I2	0755	1134	3h39	25h21
09 Dec / 343	H2H1, G1G2, F2F3, E3E2	0514	1106	5h52	31h13
10 Dec / 344	E2E1, F1F2, C2C3, D3D2	0323	0841	5h18	36h31
11 Dec / 345	C2C1, D1D2, A2A1, X5X6	0256	0751	4h55	41h26
12 Dec / 346	X3X4, S2S1, X2X1, B1B2	0237	0751	5h14	46h40
13 Dec / 347	V1V2	0334	0606	2h32	49h12
16 Dec / 350	O2O1, T3T4, X13X14, K5K56	0500	0956	4h56	54h08
17 Dec / 351	X10X9, Y1Y2, X7X8	0340	0604	2h24	56h32
Total 56h 32min					

Turbulence and mountain waves are a challenge when flying in mountainous areas. The combination of very high mountains and a strong jet stream (100+ knots from WNW) made these flights a serious challenge both in general aviation terms and especially in terms of getting useful gravity out of the flights. The flights started as early as possible in the morning in order to take advantage of the more settled conditions at the beginning of the day.

All survey line segments were flown at constant altitude. The altitude was kept as low as terrain and weather allowed.

Survey Equipment

The survey equipment was installed in Roskilde, Denmark and consisted of:

OY-CKP Beechcraft 200

LaCoste & Romberg Air/Sea gravimeter S-38

Chekan AM-25 airborne gravimeter

Javad Lexon GPS receivers (aircraft)

Javad Delta GPS receivers (aircraft and airport ground reference)

2. GPS processing and reference station coordinates.

A GPS reference station was operated in Kathmandu Airport. Coordinates for the GPS reference stations were obtained from the AUSPOS GPS service provided by Geoscience Australia. Coordinates are given in ITRF2005 frame, see <http://www.ga.gov.au/bin/gps.pl>. The AUSPOS service also provides estimates of the accuracy with which the coordinates are determined. These estimates in general amounted to 1 to 2 cm for the height component for individual sessions. Variations in the results from different sessions, typical 5 to 8 hours, were consistent with the noise estimates. Mean values for several sessions were used as final coordinates for the reference receiver, see table 2.

Aircraft trajectories were computed with the WayPoint software package from NovAtel (Calgary, Canada) using precise ephemerides from International GNSS Service (<http://igsb.jpl.nasa.gov/>). More combinations of rover and base receiver were computed for each flight and the best performing solution chosen for the final iteration of the gravity processing.

Table 2. Coordinates for airport GPS reference station (ITRF 2005)

Airport	Designation	Latitude	Longitude	Height ell. (m)
Kathmandu	KTM1	N 27 42 14.3097	E 85 21 31.4302	1297.875

3. Gravity ties

Apron reference gravity point was tied to three absolute gravity stations, one in the airport, one at the Survey Department and one in Nagarkot. A LaCoste and Romberg G-meter serial number G-466 was used for these ties.

Table 3. Apron gravity value

Airport	G value (IGSN71)	Sigma
Kathmandu airport, military apron	978664.716	0.02

4. Airborne gravity processing

Gravity results in this compilation are based solely on data from the LaCoste & Romberg instrument.

Free-air gravity anomalies at aircraft level are obtained from

$$\Delta g = f_z - f_{z0} - h'' + \delta g_{\text{eotvos}} + \delta g_{\text{tilt}} + g_0 - \gamma_0 - \left(\frac{\partial \gamma}{\partial h} (h - N) + \frac{\partial^2 \gamma}{\partial h^2} (h - N)^2 \right) \quad (1)$$

where f_z is the gravimeter observation, f_{z0} the apron base reading, h'' the GPS vertical acceleration, δg_{eotvos} the Eötvös correction computed by the formulas of Harlan (1968), g_0 the apron gravity value, γ_0 normal gravity, h the GPS ellipsoidal height and N the geoid undulation (EGM08 used throughout). γ_0 and the second order height correction in equation (1) is based on GRS80 definitions (Moritz, 1980).

An altitude dependent atmospheric correction has been applied, see eg Hintze et al 2005.

EGM08 values are derived from the pre-computed geoid undulations grid Und_min2.5x2.5_egm2008_isw=82_WGS84_TideFree_SE provided by N. Pavlis, see http://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/egm08_wgs84.html.

The platform off-level correction Δg_{tilt} is based on a platform modeling approach as described in Olesen (2003).

All data were filtered with a symmetric second order Butterworth filter with a half power point at 150 seconds, corresponding to an average spatial resolution of 6 km (half-wavelength). The spatial resolution will though differ considerably depending on flight direction as the ground speed for eastbound flights are much higher than for westbound flights due to the jet stream.

The impulse response and spectral behavior of the filter are shown in Figure 3.

Apron base readings were performed each day after the flight and approximately every second day

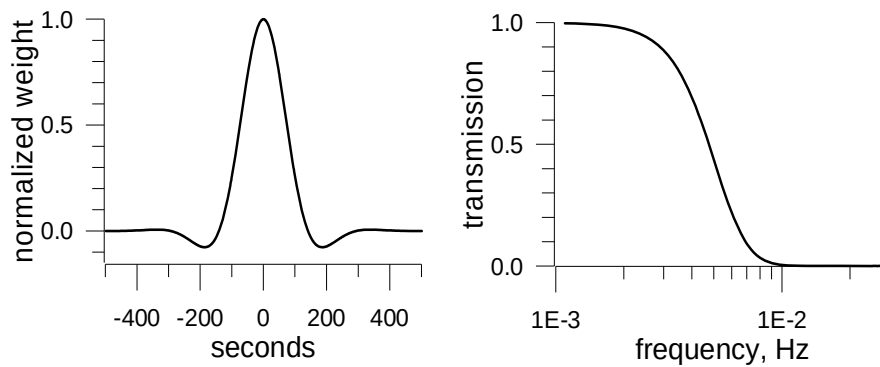


Figure 2. Impulse response (normalized) and spectral representation of the filter

before the flight in order to monitor drift of the airborne gravimeter and make a proper connection of airborne readings to the IGS71 network.

5. Data validation

Table 4. Cross over errors

	Number of line crossings	RMS
Anomalies at flight altitude	104	12.1 mGal
Anomalies reduced for terrain	104	7.3 mGal
Anomalies reduced for terrain and downward continued to 6600 m	104	4.6 mGal

The final dataset comprised 104 line intersections. Analysis of the misfit at the crossing points showed a RMS value of 12.1 mGal. The misfit is mainly due to variations in altitude, many lines are as much as 5 km apart. The value reduces to 4.6 mGal after the the effect of terrain were removed and the data were downward continued to a common level of 6600 m. 4.6 mGal cross over error indicates a 3.2 mGal noise level on individual survey lines assuming the noise to be uncorrelated from track to track.

This noise level is in line with results from other surveys over mountainous regions.

Resulting free air anomalies are pictured in Figure 3. The absence of trackiness in the plot signals a healthy dataset with little or no internal biases.

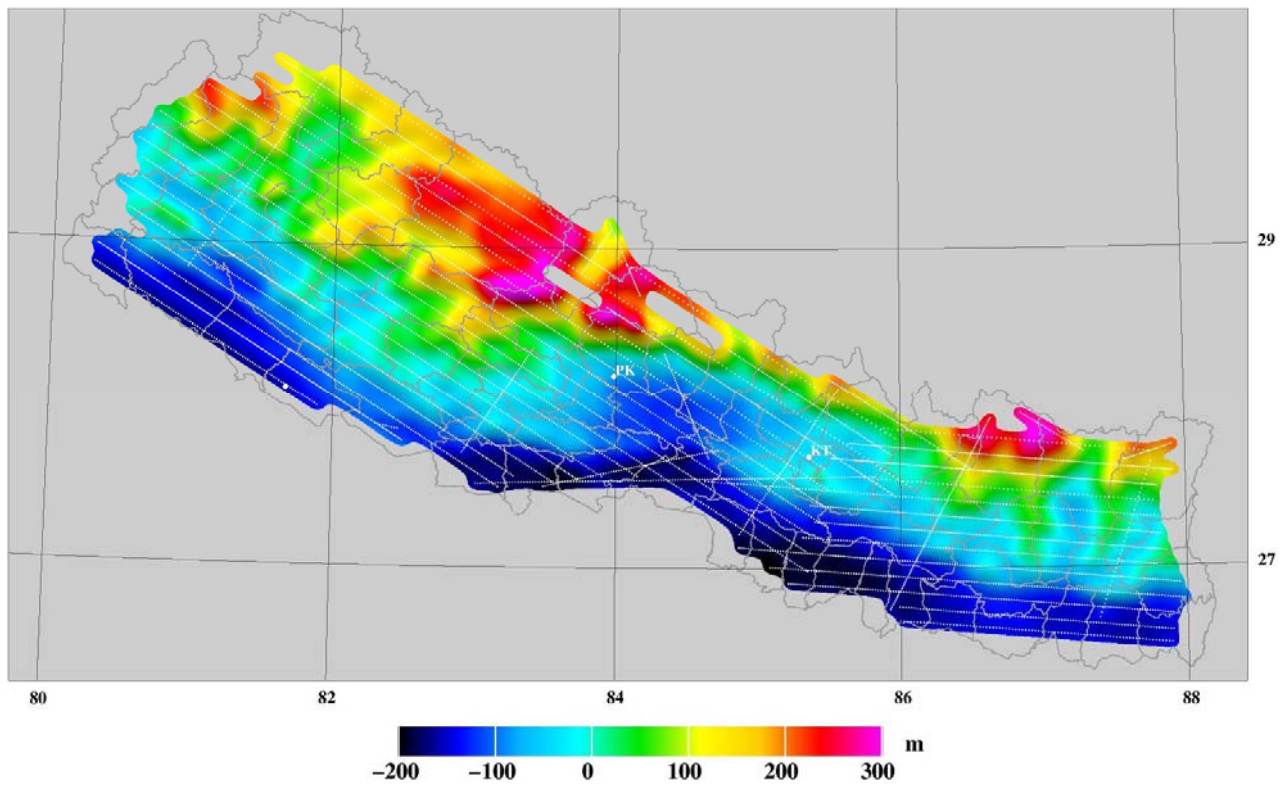


Figure 3. Free air anomaly field at flight altitude.

6. Summary

The airborne gravity survey was successfully completed in one of the most challenging settings in the world for airborne gravity measurements. The resulting free air anomalies comes with an internal noise estimate of 3.2 mGal, which is seen as very satisfactory given the challenging conditions. There is little or no sign of internal biases in the dataset.

References

- Harlan, R. B.: Eotvos corrections for airborne gravimetry, *J. Geophys. Res.*, 3, 4675-4679, 1968.
- Moritz H.: *Geodetic Reference System 1980*. Journal of Geodesy, **54**, 395-405
- Olesen, A. V.: *Improved airborne scalar gravimetry for regional gravity field mapping and geoid determination*, Kort og Matrikelstyrelsen Technical Report no. 24, 55 pp, 2003

Appendix A. Format of data file

File name: Nepal-2010.faa

Text file with one line for each data point.

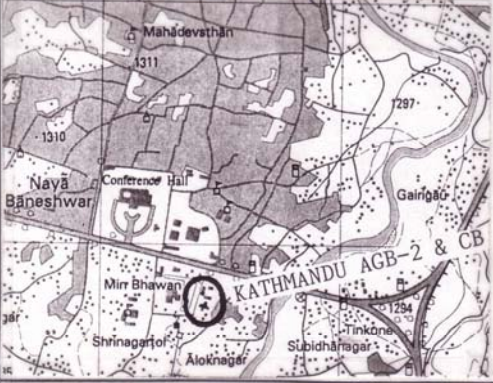
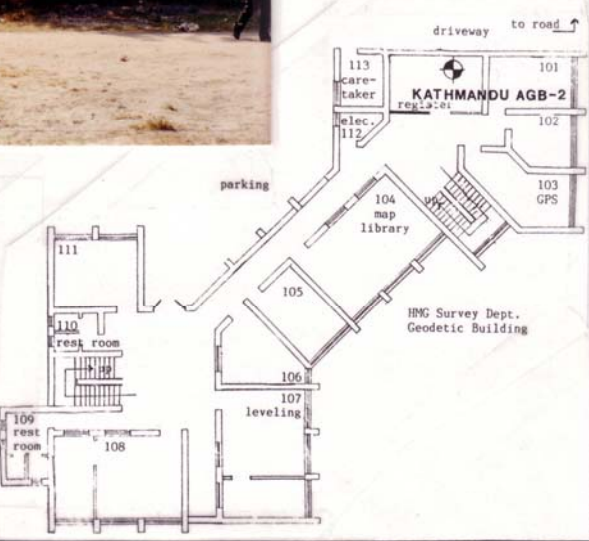
Example:

```
2111980 -1.00062 117.99714 3216.55 977066.2 25.3 306.29875
Id      Lat      Lon      H      g      FAA  Time
```

where

Id	lineID*10000+seconds from start of line
Lat	Latitude in decimal degrees
Lon	Longitude in decimal degrees
H	Orthometric height, referred to EGM08 geoid
g	Absolute gravity value in mGal
FAA	Free-air anomaly in mGal. Height dependent atmospheric correction applied
Time	Time in units of decimal-days of the year

Appendix B. Gravity reference stations

GRAVITY STATION DESCRIPTION	STATION TYPE	STATION DESIGNATION
	Absolute Station	KATHMANDU AGB-2
COUNTRY Nepal	ZONE (ANCHAL) - DISTRICT (ZILA) Bagmati - Kathmandu	CITY Kathmandu
LATITUDE 27° 41' 08.5" N	LONGITUDE 85° 20' 34.7" E	ELEVATION 1306.7 meters
GRAVITY STATION MARK 2 cm brass plug	AGENCY NOAA-NGS & HMG Survey Dep	INSCRIPTION KATH AGB2 1996
POSITION/ELEVATION REFERENCE Topo map /scaled from BM	POSITION/ELEVATION SOURCE HMG Survey Dept.	SOURCE DESIGNATION Sheet 2785 06A 1:25 000
POSITION/ELEVATION REMARKS 1.1 m below BM 102-271.2		GRAVITY VALUE/SOURCE/DATE g = 978 663.230 ± 0.003 mGal (NOAA,12/96)
DESCRIPTION/CONTACT Station is at the HMG Survey Dept. Geodetic Section Bldg. in Min Bhavan, New Baneshwar section of Kathmandu. Station is 2.3 km ESE of Maitighar Triangle or 0.9 km WNW of Tinkune Triangle and 0.15 km SSW of road. It is 3.9 km SE of Narayanhiti Royal Palace. Station is in Registrar's Room (central of 3 rooms on north face of Geodetic Bldg.), on ground level, 2.4 m N of S wall and 3.45 m E of W wall of 3.7 m by 5.65 m room, 1.3 m SSE of S side of pillar in middle of North glass-doored wall, 5.0 m SSW of 0.4 m diameter tree and 8.6 m E of and 1.4 m S of NE corner of bldg. Plug is epoxyed flush into concrete floor. Contact is Narayan Bhattarai at telephone 977-482314.		
OTHER STATION DESIGNATIONS: <i>NN 1105</i>		
DIAGRAM/PHOTOGRAPH  N ↑  		
DESCRIPTION BY Daniel Winester	AGENCY NOAA-NOS-NGS	DATE 7 January 1997

GRAVITY STATION DESCRIPTION		STATION TYPE	STATION DESIGNATION
		National Reference Base	KATHMANDU J
COUNTRY	Nepal	ZONE (ANCHAL) - DISTRICT (ZILA)	CITY
		Bagmati - Kathmandu	Kathmandu
LATITUDE	27° 41' 49.758" N	LONGITUDE	ELEVATION
		85° 21' 28.505" E	1332.006 meters
GRAVITY STATION MARK	30 mm steel hub	AGENCY	INSCRIPTION
		HMG Survey Dept.	--
POSITION/ELEVATION REFERENCE	Disk posit/disk elevation	POSITION/ELEVATION SOURCE	SOURCE DESIGNATION
		Univ of Colo/Survey Dept.	--
POSITION/ELEVATION REMARKS		GRAVITY VALUE/SOURCE/DATE	
WGS 84 GPS posit/Indian MSL 1st order lev		g = 978 661.319 ± 0.010 mGal (NOAA, 4/91)	
DESCRIPTION/CONTACT			
Station is at Trebhuvan International Airport in Kathmandu. It is 4.3 km ESE of Narayanhiti Royal Palace. To access from Tinkune Triangle, go NNE of Mean-moven Road for 2.0 km. Turn E (right) into airport and go 0.3 km to Pass Office under Control Tower & get field pass. Go S for 0.3 km, passed Int'l Terminal, to gate to E and airfield. Go 0.2 km along S edge of jet parking apron to access road to SSW. Station is about 62 m SSW of apron, 16 m WNW of WNW edge of taxiway, 10.5 m ESE of center of access road and in center of 3 m by 3 m macadam area surrounded by a white metal fence. Station is at center of 0.70 m by 0.70 m by 0.36 m deep concrete pit, 0.32 m SW of Reference Mark and 1.5 m ESE of a witness sign. Hub is set in concrete and rises 25 mm above concrete.			
OTHER STATION DESIGNATIONS: International Gravity Station, , NN 1102			
DIAGRAM/PHOTOGRAPH			
DESCRIPTION BY		AGENCY	DATE
Daniel Winester		NOAA-NOS-NGS	4 November 1996

NEPAL GRAVITY SURVEYS 1981-84

DESCRIPTIONS OF GRAVITY BASE STATIONS

BAGLUN J	KATHMANDU C
BARTUNG E	KATHMANDU CA
BENI D	* KATHMANDU CB
BHADRAPUR D	KATHMANDU CC
* BHAIRAWA D	KATHMANDU D
* BHAIRAWA J	KATHMANDU DA
BHARATPUR E	KATHMANDU DB
BIRGANJ C	KATHMANDU F
* BIRATNAGAR J	* KATHMANDU J
BUTWAL D	KATHMANDU JA
BUTWAL E	KATHMANDU K
CHARIKOT C	LAMBI E
CHAURJAHARI J	LAMOSANGU E
CHAURJAHARI JA	LAUKI E
DHALKE E	MUGLIN D
DHANGADHI C	NABALPUR E
DHANGADHI J	NAGARKOT C
DHARAN F	NEPALGANJ J
DOTI J	NEPALGANJ JA
HETAUDA D	POKHARA F
JANAKPUR J	POKHARA J
JUMLA J	POKHARA JA
KANCHANPUR E	RAPTI RIVER E
~ KODARI E	REUBEN E RUPANI E
	* SIMRA J = SIMARA
	SURKHET J = Birmandranagar
	TAULIHAWA E
	TUMLINGTAR J

→ network base station
- fixed to IGSN-71 sites
BANGKOK A + COLUMBO J

1996 Absolute gravimeter observations

* relative tie to 1981-84 network

- each absolute site has a relative meter eccentricity

NAGARKOT AAB-1

KATHMANDU AAB-2

CHAKU AAB-3

KODARI AAB-4

JIRI AAB-5

TAUSEN AAB-6 & 7

SIDDHARTHANAGAR AAB-8 (= BHAIRAWA)

BIRATNAGAR AAB-9

BHEDETAR AAB-10

SIMARA GPS

